

HNSS04 · Craniofacial Development — Timeline

Weeks post-conception, approximate · **EXAM** = flagged in class as testable · Origin key: **Arch 1** **Arches 2–4** **Occipital somites** **Endoderm** **Neural crest**
Surface ectoderm **Ultimobranchial body** **Mesoderm / mesenchyme**

HoyaDocs
 Updated Oct 1, 2026 · Rev 2

The Tongue				
WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	NERVE / CLINICAL LINK
Wk 4–5	Lateral lingual swellings + tuberculum impar	Arch 1 forms two lateral lingual swellings and one medial tuberculum impar . Underlying mesoderm + NC signal the overlying endoderm to proliferate	Arch 1	Anterior 2/3: V3 (sensation) + VII chorda tympani (taste)
Wk 5–10	Copula (hypobranchial eminence)	A second medial swelling, the copula (anterior hypobranchial eminence), forms from mesoderm under arches 2, 3 and 4 → posterior 1/3 (root)	Arches 2–4	Posterior 1/3: IX · extreme posterior: X (superior laryngeal)
Wk 5–10	Anterior 2/3 forms	Lateral swellings overgrow the tuberculum impar and merge → anterior two-thirds of the tongue body. The foramen cecum sits between the tuberculum impar and the hypobranchial eminence	Arch 1 Endoderm	Foramen cecum = where the thyroid starts
Wk 5–10	Tongue muscles	Mesoderm-derived myoblasts migrate from the occipital somites (paraxial mesoderm) and populate the tongue → tongue = mesoderm surrounded by endoderm . Same theme as the lung: mesenchyme signals endoderm	Occipital somites	XII (palatoglossus = X); XII lesion → tongue deviates toward the lesion STEP 1
Wk 5–10	Epiglottis + glottis	Posterior hypobranchial eminence (arch 4) + epiglottal swellings → epiglottis and the primordial glottis , a solid mass whose thin covering undergoes apoptosis → laryngotracheal orifice	Arches 2–4	Epiglottis = the flap; glottis = the opening (vocal-fold level)

★ **USMLE high-yield:** Front is **arch 1** (V3 + VII taste), back is **arch 3** (IX), very back is **arch 4** (X), muscles are **occipital somites** (XII). Gag reflex: afferent IX, efferent X.

The Thyroid Gland				
WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	CLINICAL LINK
~Wk 4	Thyroid primordium	Endodermal epithelium proliferates at the base of the pharynx at the foramen cecum , between the tuberculum impar and hypobranchial eminence. Forms “much like the lungs”	Endoderm	Fails to descend → lingual thyroid STEP 1
Wk 4–6	Descent on the thyroglossal duct	Descends as a bilobed diverticulum (buds only twice, unlike the lung), attached to the tongue by the thyroglossal duct , which later disappears by apoptosis . BMPs at the tip of the descending gland. PAX (an NC marker) in the thyroid → NC contributes to the thyroid EXAM . Also TTF-1, noggin	Endoderm	Duct remnant → thyroglossal duct cyst / fistula ; pyramidal lobe
Wk 6	In front of hyoid + larynx	Thyroid lies in front of the hyoid bone and laryngeal cartilages	Endoderm	Cysts often sit just below the hyoid
Wk 7	Final position	Reaches its final position in front of (overlying) the trachea . The ultimobranchial body (pouch 5/4) joins and supplies the C cells	Endoderm Ultimobranchial body	C cells → medullary thyroid carcinoma STEP 1
Wk 14–15	Thyroid starts functioning	Follicular cells make colloid → T4 + T3 (cellular metabolism). C cells secrete calcitonin (32-amino-acid peptide) → ↑ Ca ²⁺ uptake into bone, prevents resorption; fetal levels 2–3× maternal	Endoderm Ultimobranchial body	Congenital hypothyroidism (cretinism) → newborn screening STEP 1

★ **USMLE high-yield:** “Born in the tongue, walks down the neck”: foramen cecum → front of hyoid/larynx (**wk 6**) → front of trachea (**wk 7**) → switches on (**wk 14–15**). Midline neck mass that **moves with swallowing** = thyroglossal duct cyst.

The Face — Prominences, Lip, Nose, Palate, Nasal Cavity, Teeth

WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	CLINICAL LINK
~Wk 2–3.5	Neural crest into arches 1–2	NC cells migrate from rhombomeres R2 and R4 into arches 1 and 2. NC forms virtually all facial bones (exceptions: thyroid + cricoid cartilages; ? malleus/incus)	Neural crest	Tcof-1 mutation → migration fails → Treacher Collins
Wk 4–5	Five facial prominences	Frontonasal (between the 1st arches; mesoderm + NC inside, ectoderm outside; ectoderm–mesenchyme crosstalk) · paired maxillary (lateral to the stomodeum) · paired mandibular (caudal to maxillary)	Neural crest Mesoderm Surface ectoderm	—
Wk 4–5	Nasal placodes	Local thickenings of surface ectoderm on either side of the frontal prominence	Surface ectoderm	—
Wk 5–8	Nasal pits → nasal prominences	Placodes invaginate → nasal pits ; tissue ridges around each pit → medial and lateral nasal prominences	Surface ectoderm Neural crest	—
Wk 5–8	Upper lip	Maxillary prominences grow medially , pushing the nasal prominences to the midline; the groove between medial nasal + maxillary disappears and they fuse . Upper lip = 2 medial nasal + 2 maxillary (lateral nasal does not form lip)	Neural crest	Failed fusion → cleft lip (~1/1,000, 80% male)
Wk 5–8	Nose	Formed from 5 prominences : frontonasal → bridge ; 2 merged medial nasal → crest + tip ; 2 lateral nasal → sides	Neural crest	Medial nasal fail to merge → median cleft / bifid nose
Wk 5–8	Intermaxillary segment → primary palate	Deep fusion beneath the lip forms the intermaxillary segment : labial → philtrum · upper jaw → 4 incisors · palatal → triangular primary palate . Continuous with the rostral nasal septum	Neural crest	—
Wk 6	Nasal pits deepen · dental lamina	Nasal pits grow into surrounding NC and mesoderm. Basal oral epithelium forms the C-shaped dental lamina → 10 dental buds per jaw → cap stage (teeth: not tested in class)	Surface ectoderm	—
Wk 6.5–7	Palatine shelves → secondary palate	Palatine shelves grow from the inside of the maxillary prominences , toward each other dorsal to (above) the tongue , and fuse → secondary (definitive) palate = hard + soft palate including the uvula	Neural crest	Failed fusion → cleft palate ± cleft uvula; oblique facial cleft (maxillary + lateral nasal)
Wk 7	Oronasal membrane → choana	By week 7 the oronasal membrane separates the nasal pits from the mouth; it ruptures by mid-week 7 → primitive choana , joining nasal and oral cavities behind the primary palate	Surface ectoderm	—
Wk 8–9	Cheeks	Maxillary prominences have grown enough that cheeks are identifiable ; cartilage that will ossify into the maxillae is present	Neural crest	—
Wk 8–15 → birth	Teeth	Milk teeth form first; secondary (permanent) buds begin wk 8–15; tooth stages continue wk 15–20 and wk 20–birth (not tested in class)	Surface ectoderm Neural crest	—
→ Puberty	Paranasal sinuses	Diverticula of the lateral nasal wall into the maxilla, ethmoid, frontal, sphenoid ; increase surface area to condition air; keep growing through puberty and shape the face	Surface ectoderm	—

Sequence hook: Crest arrives → **5 prominences** → **pits** → **lip + nose fuse** → **shelves** close the palate over the tongue → membrane ruptures → **cheeks**. “Lip = the two M’s (Medial nasal + Maxillary).”

★ **USMLE high-yield:** **Cleft lip** = maxillary + medial nasal (primary palate); **cleft palate** = palatine shelves; separate causes, often together. **All facial clefts can result from improper NC migration/function** **EXAM**. Neurocranium = mostly **endochondral** (mesoderm + NC); viscerocranium = mostly **membranous** (NC). Down syndrome on ultrasound: **absent nasal bone**, abnormal ear position.